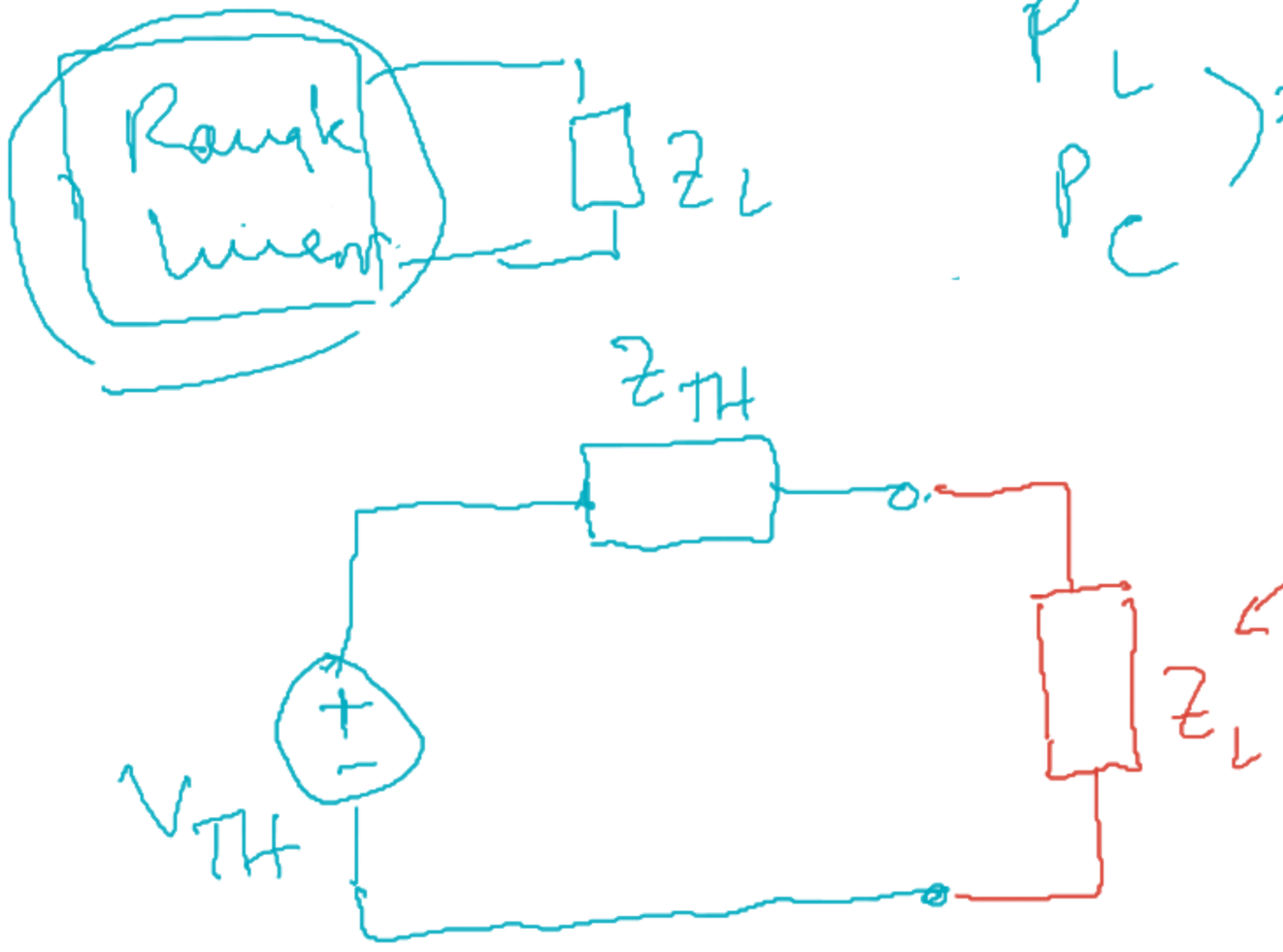



$P_L \geq 0$

$P_C$

$Z_L = Z_{TH}$

$= Z_{TH}^*$

$P_{max} = \frac{|V_{TH}|^2}{8 R_{TH}}$

$Z_{TH} = R_{TH} \pm jX_{TH}$

$$V_{rms} = \frac{1}{\sqrt{2}} V_{max}$$

$$I_{rms} = \frac{1}{\sqrt{2}} I_{max}$$

$$\Big\} \quad P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

$$x = \theta_v - \theta_i$$

$$V_m = \sqrt{2} V_{rms}$$

$$I_m = \sqrt{2} I_{rms}$$

$$\Big\} \quad P = \frac{1}{2}(\sqrt{2} V_{rms} \times \sqrt{2} I_{rms}) \cos x$$

$$= \underbrace{V_{rms} I_{rms}}_{S} \underbrace{\cos(\theta_v - \theta_i)}_{\phi}$$

$\Rightarrow$ Daya Semu (VA)

$P = \underline{S} = V_{rms} I_{rms} \Rightarrow$ Daya Semu (VA)

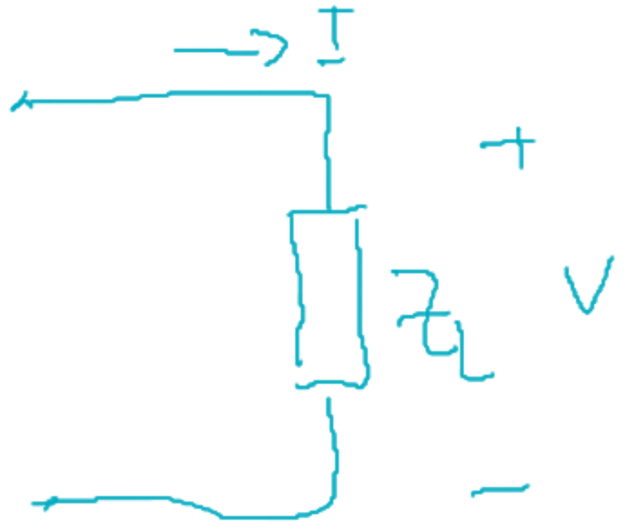
$f = \cos(\theta_v - \theta_i) \Rightarrow$ Faktor Daya

$= \cos \underline{\phi}$

$0 < f \leq 1 \Rightarrow$

Beban $\underline{L} \Rightarrow$ HK / Kerlbus

Example:




$$i(t) = 4\cos(100\pi t + 10^\circ)\ A \qquad (\omega t)$$

$$v(t) = 120\cos(100\pi t - 20^\circ)\ V$$

$S = ?$ & $pf = ?$

$I = 4\angle 10^\circ\ A \qquad V = 120\angle -20^\circ\ V$

$I_m = 4\ A$

$V_m = 120\ V$

$$S = V_{rms} \times I_{rms} = \frac{1}{\sqrt{2}} \times 120 \times \frac{1}{\sqrt{2}} \times 4$$

$$= 480/2 = 240\ VA$$

$$PF = \cos(\theta_v - \theta_i) = \cos(-20 - 10)$$

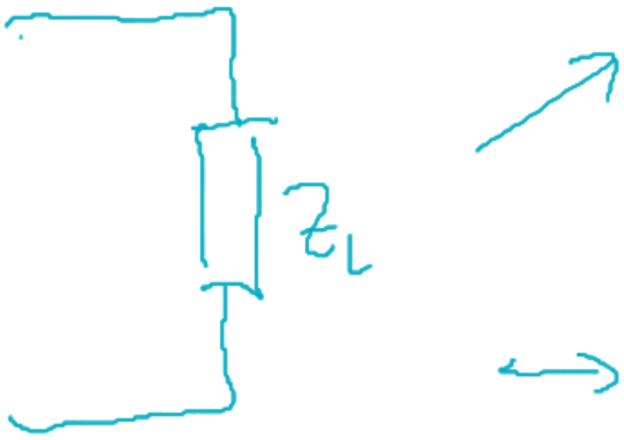
$$= \cos -30^\circ$$

$$= \cos 30^\circ =$$

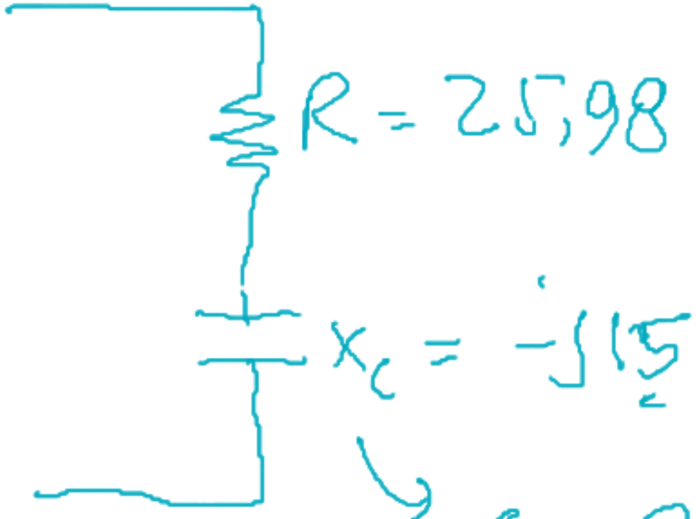
$$= 0,\cancel{5}4$$

$$= 0,866$$

$$Z_L = \frac{V}{I} = 25{,}98 - j15\ \Omega$$




$\leftrightarrow$ $Z_L$




$$X_C = \frac{1}{\omega C} \Rightarrow \omega = 100\pi \Rightarrow \pi = 3{,}14$$

$$C = \frac{1}{X_C \cdot \omega} = \frac{1}{15 \times 100 \times 3{,}14} = \frac{1}{4710} = 0{,}000212\ F$$

$$C = 0{,}0002123\ F$$

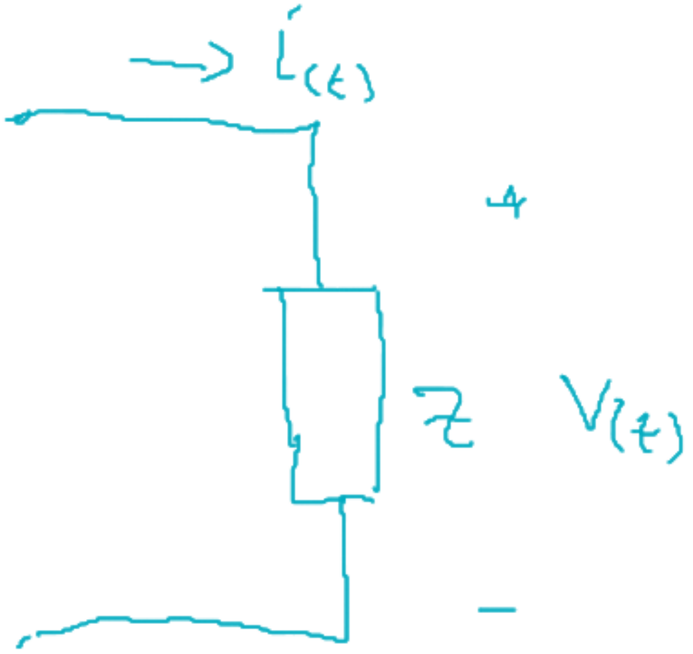
$$= 212{,}3\ 10^{-6}\ F$$

$$= 212{,}3\ \mu F$$

Contoh :

$Z = 60 + j40\ \Omega$

$V_{(t)} = 320 \cos(377t + 10^\circ)\ V$




S = ? & pF = ?

$V = 320 \angle 10^\circ\ V$.

$Z = \sqrt{3600 + 1600} \angle \tan^{-1} \frac{40}{60}$

$= 72,1 \angle 33,69^\circ$

$$\underline{I} = \frac{320\angle 10}{72{,}1\angle 33{,}69^\circ} = 4{,}43\angle -23{,}69^\circ \quad \leftarrow \underline{I}_m$$

$$S = V_{rms} \times I_{rms} = \frac{1}{2} V_m \cdot I_m$$

$$= \frac{320}{\sqrt{2}} \times \frac{4{,}43}{\sqrt{2}} = 708{,}8 \text{ VA}$$

$$PF = \cos(10 + 23{,}69) = 0{,}832$$

DAYA AC → * Daya rata-rata ⇒ P = Watt

↓ * Daya sesaat ⇒ P(t) =

Nilai RMS * Daya semu - S = VA

= * Faktor Daya - PF ≠

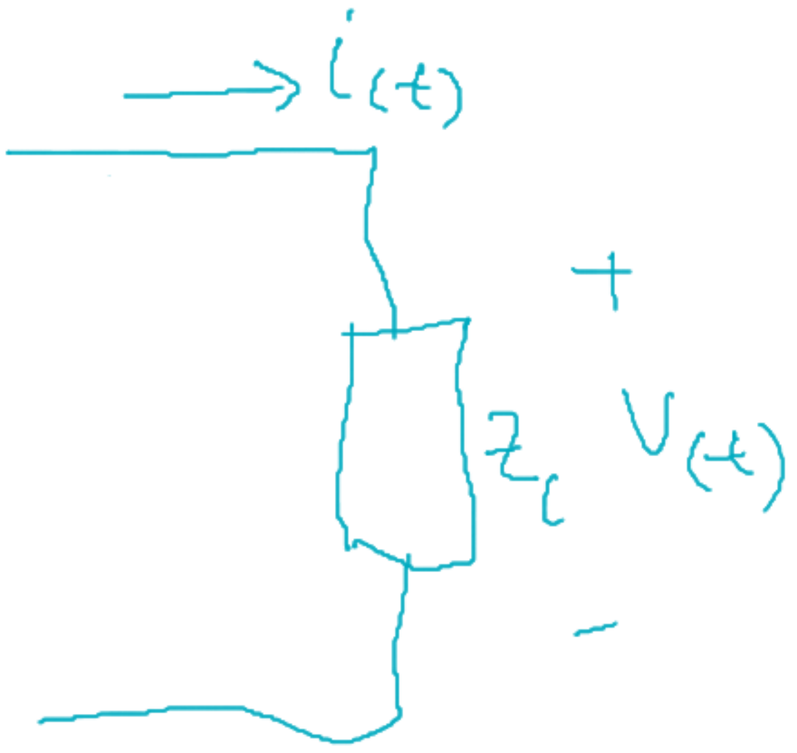
* Daya Reaktif - Q = VAR

$$S = P + jQ$$

VA  W  VAR

$$P = Re(S) = I_{rms}^2 R$$

$$Q = Im(S) = I_{rms}^2 X$$




$P = 22,5$ watt

$Q = -38,97$ VAR

$P = ?$

$S = ?$

$Q = ?$

$PF = ?$

$Z = ?$

$S = 45 \angle -60^\circ$

$S = P + jQ$

$P = 45 \cos 60$

$Q = 45 \sin -60$

$= -45 \sin 60$

R

$R + jX$

$R - jX$